EXP 14	Overdue Identification and Library Dashboard

Aim:

To identify overdue books and to build a dashboard showing the key library figures.

Theory:

- **Overdue rule:** A book is overdue when it has not been returned and its due date is before today. The status is set to Overdue when an issue is saved or edited after the due date, and the function markOverdueIssues() updates any Issued record whose due date has passed.
- **Dashboard page:** Built with the Page Builder (ZML). Panels show Total Book Titles, Total Members, Total Available Copies, Books Currently Issued and Overdue Books; a pie chart shows issues by status, a column chart shows available copies by category and the Currently Issued report is embedded.

Procedure:

1. Create the function markOverdueIssues() in Workflow > Functions and execute it.
2. Click + > Page > Blank, name it Library Dashboard.
3. Open the page in the Application IDE and write the ZML with five panels, two charts and the embedded report.
4. Save and open Library Dashboard in the live application.

Program (markOverdueIssues function):

```
void markOverdueIssues()
{
    // mark every unreturned issue whose due date has passed as Overdue
    count = 0;
    for each rec in Book_Issues[Issue_Status == "Issued" && Due_Date <
zoho.currentdate]
    {
        rec.Issue_Status = "Overdue";
        count = count + 1;
    }
    info "Issues marked Overdue: " + count;
}
```

Program (dashboard panels - ZML extract):

```
<panel elementName='Total_Titles_KPI'> ... <text type='Form Data'
value='thisapp.Books.ID.count'> ... </panel>
<panel elementName='Total_Members_KPI'> ... value='thisapp.Members.ID.count' ...
</panel>
<panel elementName='Available_Copies_KPI'> ...
value='thisapp.Books.Available_Copies.sum' ... </panel>
<panel elementName='Currently_Issued_KPI'> ... value='thisapp.Book_Issues.ID.count'
       criteria='(Issue_Status != "Returned")' ... </panel>
<panel elementName='Overdue_KPI'> ... value='thisapp.Book_Issues.ID.count'
       criteria='(Issue_Status == "Overdue")' ... </panel>
<chart elementName='Issues_by_Status' type='Pie' formLinkName='Book_Issues'
       xfield='Issue_Status' yfields='count:Issue_ID' />
<chart elementName='Available_Copies_by_Category' type='Column' formLinkName='Books'
       xfield='Category' yfields='sum:Available_Copies' />
```

```
<report elementName='Currently_Issued_Report' linkName='Currently_Issued' />
```

Output:

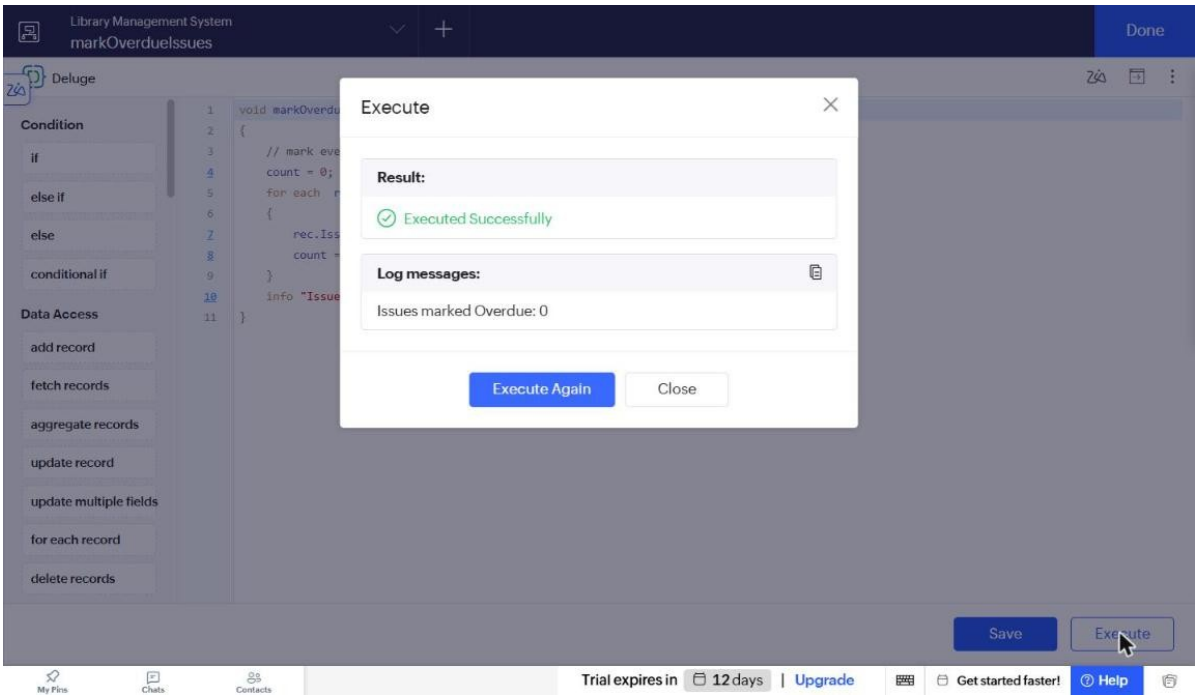


Fig. 14.1: markOverdueIssues executed - no pending Issued record past due

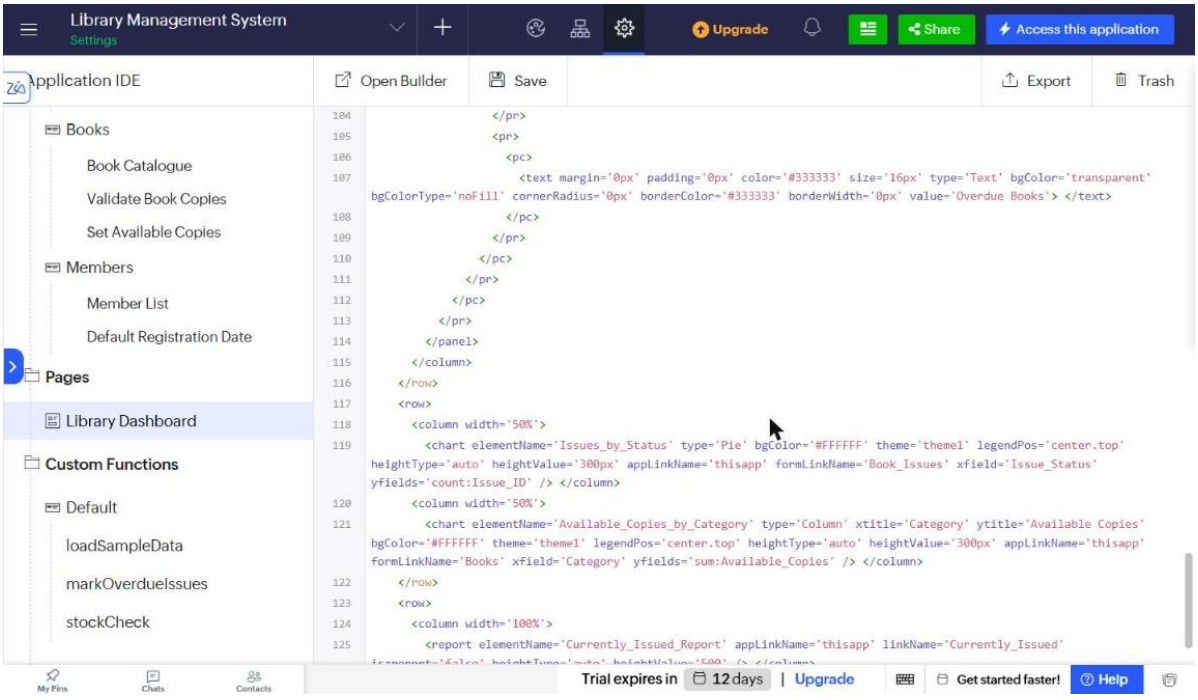


Fig. 14.2: Dashboard ZML in the Application IDE

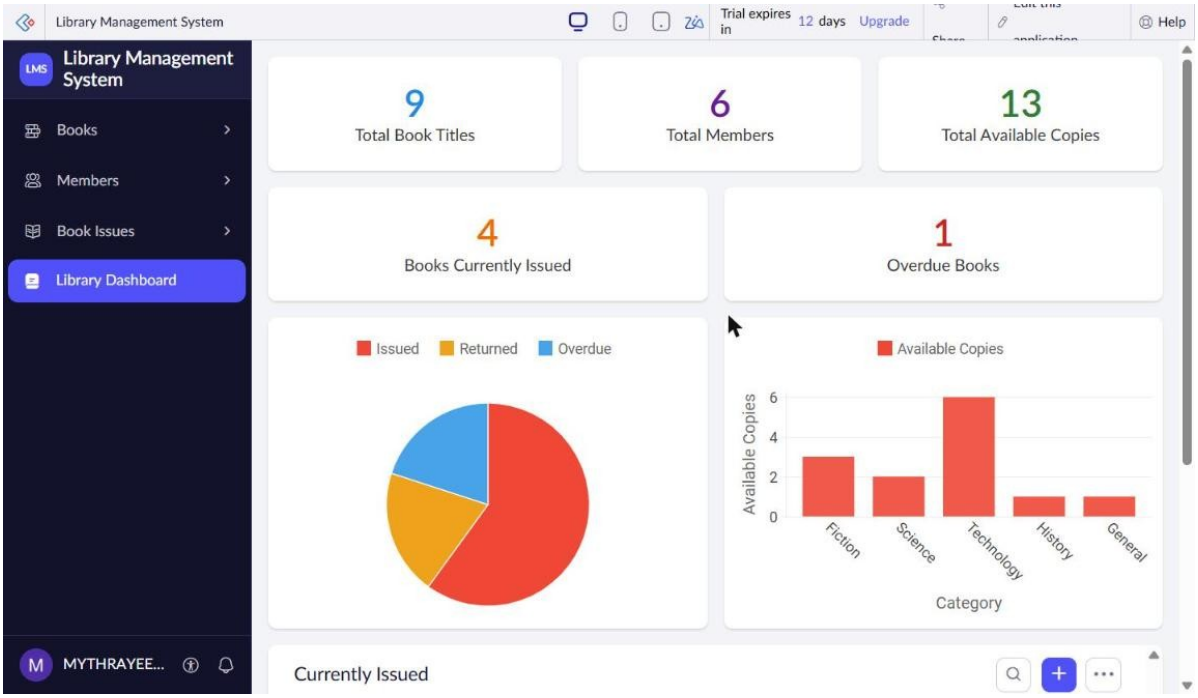


Fig. 14.3: Library Dashboard

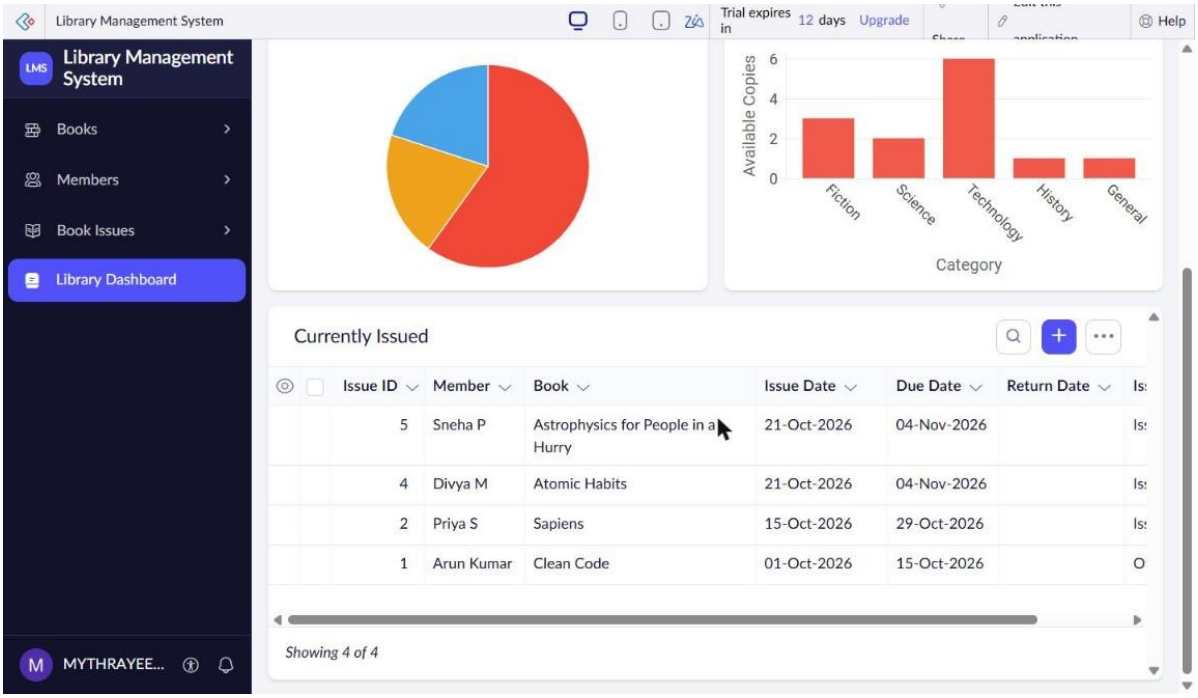


Fig. 14.4: Dashboard - embedded Currently Issued report

Observation (Dashboard on 21-Oct-2026):

Indicator	Value
Total Book Titles	9
Total Members	6
Total Available Copies	13
Books Currently Issued	4

Overdue Books	1 (Clean Code - Arun Kumar)

Result:

Overdue issues are identified automatically and the Library Dashboard shows the total titles, members, available copies, currently issued and overdue counts.